NAVYA SREERAMOJU

AI/ML Engineer |Robotics Engineer

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PROFILE

AI/ML Engineer with an M.S. in Artificial Intelligence from The University of Texas at Austin and a foundation in automation and robotics. Strong technical expertise in deep learning, neural network architectures, optimization algorithms, and model evaluation, with an emphasis on implementing ML methods from first principles and understanding their computational trade-offs. Proficient in PyTorch, transformer and vision-language models, generative modeling, parameter-efficient fine-tuning, and low-bit quantization. Experienced in AI application development, backend engineering, and inference workflows, with additional expertise in robotic perception, state estimation, and ROS 2. Focused on efficient ML systems and the engineering challenges at the intersection of machine learning and intelligent robotics.

SKILLS

- **Programming:** Python, C, C++, SQL, R, MATLAB
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, PEFT, NumPy, Pandas
- **Generative AI & LLMs:** Transformer Models, GPT, RAG, Prompt Engineering, LLM Evaluation, Diffusion Models, Rectified Flow, Energy-Based Models, WGAN-GP, Normalizing Flows
- **Model Optimization:** 4-bit Quantization, LoRA, QLoRA, Contrastive Learning, Mixed-Precision Concepts
- **Computer Vision:** OpenCV, Vision-Language Models, CLIP, Autonomous-Driving Scene Understanding
- **Software Engineering & Cloud:** FastAPI, REST APIs, Docker, Azure, Cosmos DB, Git, Data Pipelines
- **Robotics:** ROS 2 Humble, Gazebo, URDF, Robot Telemetry, Navigation Components, Kalman Filtering

AI & MACHINE LEARNING PROJECTS

Career Buddy- AI Agent & Job Intelligence Platform | Python, FastAPI, Docker, Azure, Cosmos DB, APIs

- Developing an AI-powered platform for job analysis, skill extraction, candidate-job matching, resume tailoring, and interview preparation.
- Designing modular components for API handling, business logic, agent orchestration, external tool integration, vector search, and persistent state.
- Developing state-machine-based workflows to coordinate specialized AI tasks.

AI Chatbot for FDA Drug Labeling Question Answering | Azure OpenAI, RAG, NLP, Evaluation

- Co-authored a peer-reviewed MDPI publication on source-grounded question answering for FDA drug-label documents.
- Worked with document ingestion, chunking, retrieval, and prompt grounding to generate evidence-based responses.
- Evaluated response quality using embedding-based similarity/alignment measures and ROUGE metrics.

Vision-Language Model Fine-Tuning and CLIP Implementation | PyTorch, Hugging Face, PEFT

- Fine-tuned SmolVLM-256M on a custom autonomous-driving visual question-answering dataset using LoRA with rank 8 and answer-masked labels.
- Implemented CLIP-style dual encoders, projection heads, learnable temperature, and symmetric contrastive loss.
- Evaluated batched predictions using exact-match scoring for questions involving object counting, spatial reasoning, and scene understanding.

Generative AI Models: Implementation & Optimization | PyTorch, Generative Modeling

- Implemented energy-based models using Langevin dynamics on two-dimensional Gaussian-mixture data.
- Built Wasserstein GANs with gradient penalty (WGAN-GP) and experimented with MNIST and Fashion-MNIST.
- Implemented normalizing flows, diffusion models, and rectified flow to explore different generative modeling approaches.

Analyzing and Mitigating Dataset Artifacts in Natural Language Inference | Python, PyTorch, Transformers | Python, PyTorch, ELECTRA-small, NLP

- Fine-tuned ELECTRA-small on the Stanford Natural Language Inference dataset.
- Investigated dataset shortcuts involving lexical overlap, negation, antonyms, and hypothesis-only signals.
- Built a hypothesis-only classifier and experimented with logit-subtraction debiasing to study model reliance on dataset artifacts.

ROBOTICS, PERCEPTION & SYSTEMS PROJECTS

Deep Learning Autonomous-Driving Planner | PyTorch, CNNs, Transformers

- Developed a learning-based planner to predict waypoints from visual and track-boundary inputs.
- Evaluated trajectory predictions using custom trajectory-error metrics.
- Used TensorBoard to compare trajectory smoothness and deviation across training iterations in simulation.

Real-Time Robotic Anomaly Detection with ROS 2 | PyTorch, ROS 2 Humble, Docker, RViz2

- Developing an autoencoder-based anomaly detector for robotic telemetry using a 34–16–8–16–34 neural network architecture.
- Organized the system into three ROS 2 nodes for telemetry publishing, anomaly detection, and logging.
- Containerized the project with Docker and built the ROS 2 workspace; end-to-end execution and visualization remain in progress.

Kalman Filter and LSTM Trajectory Estimation | Python, Kalman Filtering, LSTM, Time-Series Modeling

- Exploring a hybrid Kalman filter and LSTM approach for robot trajectory estimation under sensor noise and temporary occlusion.
- Investigating how traditional state estimation and sequence learning can complement each other.
- Planning evaluation using real-world odometry data.

EDUCATION

Master of Science in Artificial Intelligence
University of Texas at Austin, USA - Graduation: 2026, CGPA: 3.76/4.0

Master of Engineering in Automation and Robotics
Osmania University College of Engineering, Hyderabad-India, GPA: 8.67/10

Bachelor of Technology in Mechanical Engineering
SR Engineering College, Warangal-India,

WORK EXPERIENCE

Teaching Assistant- Automation and Robotics Lab
Osmania University College of Engineering, Hyderabad, India | August 2018 - April 2019

PROCESS PIPING DESIGN ENGINEER
Groworld Tecknica Engineering Private Limited | January 2019 - June 2020

PROCESS ENGINEER- PIPING ENGINEERING
Anewa Engineering Private Limited, Hyderabad, India | February 2015 - October 2016

PUBLICATION

Co-author — Peer-Reviewed MDPI Publication: AI Chatbot for FDA Drug Labeling Question Answering
Contributed to a document-grounded question-answering system using Azure OpenAI and retrieval-based workflows; evaluated answer alignment against reference responses.